# Model Architecture: Llama-3.1-8B-Instruct

Generated by sg_modelload

July 21, 2026

## 1 Model Overview

| | |
|---|---|
| Model Name | Llama-3.1-8B-Instruct |
| Path | /models/Llama-3.1-8B-Instruct |
| Total Parameters | 8,030,261,248 |
| Weight Dtype | bfloat16 |

## 2 Forward Pass Algorithm

**Algorithm 1** Forward pass of Llama-3.1-8B-Instruct

$h \leftarrow \text{embed_tokens}(x) \in \mathbb{R}^{B \times T \times 4096}$
**for** $layer \leftarrow 0$ **to** 31 **do**
    $h \leftarrow \text{layer}_{layer}(h)$
**end for**
$h \leftarrow \text{norm}(h) \in \mathbb{R}^{B \times T \times 4096}$
$\text{logits} \leftarrow \text{lm_head}(h) \in \mathbb{R}^{B \times T \times 128256}$

## 3 Layer-by-Layer Architecture

All weight tensors and their dimensions are listed below.

| Layer Path | Type | Weight Shape | Dtype | Params |
|---|---|---|---|---|
| model.layers.0 | LlamaDecoderLayer | – | – | – |
| model.layers.0.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.0.mlp | LlamaMLP | – | – | – |
| model.layers.0.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.0.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.0.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.0.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.0.self_attn | LlamaAttention | – | – | – |
| model.layers.0.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.0.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.0.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.0.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.1 | LlamaDecoderLayer | – | – | – |
| model.layers.1.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.1.mlp | LlamaMLP | – | – | – |
| model.layers.1.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.1.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.1.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.1.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.1.self_attn | LlamaAttention | – | – | – |
| model.layers.1.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.1.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.1.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |

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| Layer Path | Type | Weight Shape | Dtype | Params |
|---|---|---|---|---|
| model.layers.1.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.2 | LlamaDecoderLayer | – | – | – |
| model.layers.2.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.2.mlp | LlamaMLP | – | – | – |
| model.layers.2.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.2.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.2.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.2.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.2.self_attn | LlamaAttention | – | – | – |
| model.layers.2.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.2.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.2.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.2.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.3 | LlamaDecoderLayer | – | – | – |
| model.layers.3.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.3.mlp | LlamaMLP | – | – | – |
| model.layers.3.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.3.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.3.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.3.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.3.self_attn | LlamaAttention | – | – | – |
| model.layers.3.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.3.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.3.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.3.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.4 | LlamaDecoderLayer | – | – | – |
| model.layers.4.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.4.mlp | LlamaMLP | – | – | – |
| model.layers.4.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.4.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.4.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.4.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.4.self_attn | LlamaAttention | – | – | – |
| model.layers.4.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.4.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.4.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.4.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.5 | LlamaDecoderLayer | – | – | – |
| model.layers.5.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.5.mlp | LlamaMLP | – | – | – |
| model.layers.5.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.5.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.5.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.5.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.5.self_attn | LlamaAttention | – | – | – |
| model.layers.5.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.5.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.5.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.5.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.6 | LlamaDecoderLayer | – | – | – |
| model.layers.6.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.6.mlp | LlamaMLP | – | – | – |
| model.layers.6.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.6.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.6.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.6.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.6.self_attn | LlamaAttention | – | – | – |
| model.layers.6.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |

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| Layer Path | Type | Weight Shape | Dtype | Params |
|---|---|---|---|---|
| model.layers.6.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.6.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.6.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.7 | LlamaDecoderLayer | – | – | – |
| model.layers.7.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.7.mlp | LlamaMLP | – | – | – |
| model.layers.7.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.7.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.7.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.7.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.7.self_attn | LlamaAttention | – | – | – |
| model.layers.7.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.7.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.7.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.7.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.8 | LlamaDecoderLayer | – | – | – |
| model.layers.8.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.8.mlp | LlamaMLP | – | – | – |
| model.layers.8.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.8.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.8.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.8.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.8.self_attn | LlamaAttention | – | – | – |
| model.layers.8.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.8.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.8.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.8.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.9 | LlamaDecoderLayer | – | – | – |
| model.layers.9.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.9.mlp | LlamaMLP | – | – | – |
| model.layers.9.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.9.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.9.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.9.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.9.self_attn | LlamaAttention | – | – | – |
| model.layers.9.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.9.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.9.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.9.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.10 | LlamaDecoderLayer | – | – | – |
| model.layers.10.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.10.mlp | LlamaMLP | – | – | – |
| model.layers.10.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.10.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.10.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.10.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.10.self_attn | LlamaAttention | – | – | – |
| model.layers.10.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.10.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.10.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.10.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.11 | LlamaDecoderLayer | – | – | – |
| model.layers.11.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.11.mlp | LlamaMLP | – | – | – |
| model.layers.11.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.11.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.11.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.11.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |

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| Layer Path | Type | Weight Shape | Dtype | Params |
|---|---|---|---|---|
| model.layers.11.self_attn | LlamaAttention | – | – | – |
| model.layers.11.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.11.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.11.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.11.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.12 | LlamaDecoderLayer | – | – | – |
| model.layers.12.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.12.mlp | LlamaMLP | – | – | – |
| model.layers.12.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.12.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.12.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.12.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.12.self_attn | LlamaAttention | – | – | – |
| model.layers.12.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.12.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.12.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.12.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.13 | LlamaDecoderLayer | – | – | – |
| model.layers.13.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.13.mlp | LlamaMLP | – | – | – |
| model.layers.13.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.13.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.13.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.13.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.13.self_attn | LlamaAttention | – | – | – |
| model.layers.13.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.13.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.13.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.13.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.14 | LlamaDecoderLayer | – | – | – |
| model.layers.14.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.14.mlp | LlamaMLP | – | – | – |
| model.layers.14.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.14.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.14.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.14.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.14.self_attn | LlamaAttention | – | – | – |
| model.layers.14.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.14.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.14.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.14.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.15 | LlamaDecoderLayer | – | – | – |
| model.layers.15.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.15.mlp | LlamaMLP | – | – | – |
| model.layers.15.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.15.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.15.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.15.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.15.self_attn | LlamaAttention | – | – | – |
| model.layers.15.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.15.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.15.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.15.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.16 | LlamaDecoderLayer | – | – | – |
| model.layers.16.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.16.mlp | LlamaMLP | – | – | – |
| model.layers.16.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.16.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |

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| Layer Path | Type | Weight Shape | Dtype | Params |
|---|---|---|---|---|
| model.layers.16.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.16.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.16.self_attn | LlamaAttention | – | – | – |
| model.layers.16.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.16.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.16.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.16.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.17 | LlamaDecoderLayer | – | – | – |
| model.layers.17.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.17.mlp | LlamaMLP | – | – | – |
| model.layers.17.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.17.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.17.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.17.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.17.self_attn | LlamaAttention | – | – | – |
| model.layers.17.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.17.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.17.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.17.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.18 | LlamaDecoderLayer | – | – | – |
| model.layers.18.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.18.mlp | LlamaMLP | – | – | – |
| model.layers.18.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.18.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.18.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.18.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.18.self_attn | LlamaAttention | – | – | – |
| model.layers.18.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.18.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.18.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.18.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.19 | LlamaDecoderLayer | – | – | – |
| model.layers.19.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.19.mlp | LlamaMLP | – | – | – |
| model.layers.19.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.19.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.19.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.19.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.19.self_attn | LlamaAttention | – | – | – |
| model.layers.19.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.19.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.19.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.19.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.20 | LlamaDecoderLayer | – | – | – |
| model.layers.20.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.20.mlp | LlamaMLP | – | – | – |
| model.layers.20.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.20.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.20.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.20.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.20.self_attn | LlamaAttention | – | – | – |
| model.layers.20.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.20.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.20.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.20.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.21 | LlamaDecoderLayer | – | – | – |
| model.layers.21.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.21.mlp | LlamaMLP | – | – | – |

*Continued on next page*

| Layer Path | Type | Weight Shape | Dtype | Params |
|---|---|---|---|---|
| model.layers.21.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.21.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.21.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.21.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.21.self_attn | LlamaAttention | – | – | – |
| model.layers.21.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.21.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.21.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.21.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.22 | LlamaDecoderLayer | – | – | – |
| model.layers.22.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.22.mlp | LlamaMLP | – | – | – |
| model.layers.22.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.22.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.22.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.22.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.22.self_attn | LlamaAttention | – | – | – |
| model.layers.22.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.22.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.22.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.22.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.23 | LlamaDecoderLayer | – | – | – |
| model.layers.23.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.23.mlp | LlamaMLP | – | – | – |
| model.layers.23.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.23.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.23.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.23.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.23.self_attn | LlamaAttention | – | – | – |
| model.layers.23.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.23.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.23.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.23.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.24 | LlamaDecoderLayer | – | – | – |
| model.layers.24.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.24.mlp | LlamaMLP | – | – | – |
| model.layers.24.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.24.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.24.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.24.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.24.self_attn | LlamaAttention | – | – | – |
| model.layers.24.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.24.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.24.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.24.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.25 | LlamaDecoderLayer | – | – | – |
| model.layers.25.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.25.mlp | LlamaMLP | – | – | – |
| model.layers.25.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.25.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.25.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.25.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.25.self_attn | LlamaAttention | – | – | – |
| model.layers.25.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.25.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.25.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.25.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.26 | LlamaDecoderLayer | – | – | – |

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| Layer Path | Type | Weight Shape | Dtype | Params |
|---|---|---|---|---|
| model.layers.26.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.26.mlp | LlamaMLP | – | – | – |
| model.layers.26.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.26.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.26.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.26.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.26.self_attn | LlamaAttention | – | – | – |
| model.layers.26.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.26.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.26.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.26.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.27 | LlamaDecoderLayer | – | – | – |
| model.layers.27.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.27.mlp | LlamaMLP | – | – | – |
| model.layers.27.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.27.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.27.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.27.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.27.self_attn | LlamaAttention | – | – | – |
| model.layers.27.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.27.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.27.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.27.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.28 | LlamaDecoderLayer | – | – | – |
| model.layers.28.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.28.mlp | LlamaMLP | – | – | – |
| model.layers.28.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.28.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.28.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.28.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.28.self_attn | LlamaAttention | – | – | – |
| model.layers.28.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.28.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.28.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.28.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.29 | LlamaDecoderLayer | – | – | – |
| model.layers.29.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.29.mlp | LlamaMLP | – | – | – |
| model.layers.29.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.29.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.29.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.29.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.29.self_attn | LlamaAttention | – | – | – |
| model.layers.29.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.29.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.29.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.29.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.30 | LlamaDecoderLayer | – | – | – |
| model.layers.30.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.30.mlp | LlamaMLP | – | – | – |
| model.layers.30.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.30.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.30.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.30.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.30.self_attn | LlamaAttention | – | – | – |
| model.layers.30.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.30.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.30.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |

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| Layer Path | Type | Weight Shape | Dtype | Params |
|---|---|---|---|---|
| model.layers.30.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.31 | LlamaDecoderLayer | – | – | – |
| model.layers.31.input_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.31.mlp | LlamaMLP | – | – | – |
| model.layers.31.mlp.down_proj.weight | Linear | 4096 × 14336 | bfloat16 | 58,720,256 |
| model.layers.31.mlp.gate_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.31.mlp.up_proj.weight | Linear | 14336 × 4096 | bfloat16 | 58,720,256 |
| model.layers.31.post_attention_layernorm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |
| model.layers.31.self_attn | LlamaAttention | – | – | – |
| model.layers.31.self_attn.k_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| model.layers.31.self_attn.o_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.31.self_attn.q_proj.weight | Linear | 4096 × 4096 | bfloat16 | 16,777,216 |
| model.layers.31.self_attn.v_proj.weight | Linear | 1024 × 4096 | bfloat16 | 4,194,304 |
| lm_head.weight | Linear | 128256 × 4096 | bfloat16 | 525,336,576 |
| model | LlamaModel | – | – | – |
| model.embed_tokens.weight | Embedding | 128256 × 4096 | bfloat16 | 525,336,576 |
| model.layers | ModuleList | – | – | – |
| model.norm.weight | LlamaRMSNorm | 4096 | bfloat16 | 4,096 |

# 4 Parameter Count Estimation

| Component | Parameters | % of Total |
|---|---|---|
| MLP | 5,637,144,576 | 70.2% |
| Attention | 1,342,177,280 | 16.7% |
| Embedding | 525,336,576 | 6.5% |
| LM Head | 525,336,576 | 6.5% |
| LayerNorm | 266,240 | 0.0% |
| **Total** | **8,030,261,248** | **100.0%** |

### 4.1 Per-Layer Parameter Count

| Layer | Params | Running Total |
|---|---|---|
| model.layers.0.input_layernorm | 4,096 | 4,096 |
| model.layers.0.mlp.down_proj | 58,720,256 | 58,724,352 |
| model.layers.0.mlp.gate_proj | 58,720,256 | 117,444,608 |
| model.layers.0.mlp.up_proj | 58,720,256 | 176,164,864 |
| model.layers.0.post_attention_layernorm | 4,096 | 176,168,960 |
| model.layers.0.self_attn.k_proj | 4,194,304 | 180,363,264 |
| model.layers.0.self_attn.o_proj | 16,777,216 | 197,140,480 |
| model.layers.0.self_attn.q_proj | 16,777,216 | 213,917,696 |
| model.layers.0.self_attn.v_proj | 4,194,304 | 218,112,000 |
| model.layers.1.input_layernorm | 4,096 | 218,116,096 |
| model.layers.1.mlp.down_proj | 58,720,256 | 276,836,352 |
| model.layers.1.mlp.gate_proj | 58,720,256 | 335,556,608 |
| model.layers.1.mlp.up_proj | 58,720,256 | 394,276,864 |
| model.layers.1.post_attention_layernorm | 4,096 | 394,280,960 |
| model.layers.1.self_attn.k_proj | 4,194,304 | 398,475,264 |
| model.layers.1.self_attn.o_proj | 16,777,216 | 415,252,480 |
| model.layers.1.self_attn.q_proj | 16,777,216 | 432,029,696 |
| model.layers.1.self_attn.v_proj | 4,194,304 | 436,224,000 |
| model.layers.2.input_layernorm | 4,096 | 436,228,096 |
| model.layers.2.mlp.down_proj | 58,720,256 | 494,948,352 |
| model.layers.2.mlp.gate_proj | 58,720,256 | 553,668,608 |
| model.layers.2.mlp.up_proj | 58,720,256 | 612,388,864 |
| model.layers.2.post_attention_layernorm | 4,096 | 612,392,960 |
| model.layers.2.self_attn.k_proj | 4,194,304 | 616,587,264 |
| model.layers.2.self_attn.o_proj | 16,777,216 | 633,364,480 |
| model.layers.2.self_attn.q_proj | 16,777,216 | 650,141,696 |
| model.layers.2.self_attn.v_proj | 4,194,304 | 654,336,000 |
| model.layers.3.input_layernorm | 4,096 | 654,340,096 |
| model.layers.3.mlp.down_proj | 58,720,256 | 713,060,352 |
| model.layers.3.mlp.gate_proj | 58,720,256 | 771,780,608 |
| model.layers.3.mlp.up_proj | 58,720,256 | 830,500,864 |
| model.layers.3.post_attention_layernorm | 4,096 | 830,504,960 |
| model.layers.3.self_attn.k_proj | 4,194,304 | 834,699,264 |
| model.layers.3.self_attn.o_proj | 16,777,216 | 851,476,480 |
| model.layers.3.self_attn.q_proj | 16,777,216 | 868,253,696 |
| model.layers.3.self_attn.v_proj | 4,194,304 | 872,448,000 |
| model.layers.4.input_layernorm | 4,096 | 872,452,096 |
| model.layers.4.mlp.down_proj | 58,720,256 | 931,172,352 |
| model.layers.4.mlp.gate_proj | 58,720,256 | 989,892,608 |
| model.layers.4.mlp.up_proj | 58,720,256 | 1,048,612,864 |
| model.layers.4.post_attention_layernorm | 4,096 | 1,048,616,960 |
| model.layers.4.self_attn.k_proj | 4,194,304 | 1,052,811,264 |
| model.layers.4.self_attn.o_proj | 16,777,216 | 1,069,588,480 |
| model.layers.4.self_attn.q_proj | 16,777,216 | 1,086,365,696 |
| model.layers.4.self_attn.v_proj | 4,194,304 | 1,090,560,000 |
| model.layers.5.input_layernorm | 4,096 | 1,090,564,096 |
| model.layers.5.mlp.down_proj | 58,720,256 | 1,149,284,352 |
| model.layers.5.mlp.gate_proj | 58,720,256 | 1,208,004,608 |
| model.layers.5.mlp.up_proj | 58,720,256 | 1,266,724,864 |
| model.layers.5.post_attention_layernorm | 4,096 | 1,266,728,960 |
| model.layers.5.self_attn.k_proj | 4,194,304 | 1,270,923,264 |
| model.layers.5.self_attn.o_proj | 16,777,216 | 1,287,700,480 |
| model.layers.5.self_attn.q_proj | 16,777,216 | 1,304,477,696 |
| model.layers.5.self_attn.v_proj | 4,194,304 | 1,308,672,000 |
| model.layers.6.input_layernorm | 4,096 | 1,308,676,096 |
| model.layers.6.mlp.down_proj | 58,720,256 | 1,367,396,352 |
| model.layers.6.mlp.gate_proj | 58,720,256 | 1,426,116,608 |
| model.layers.6.mlp.up_proj | 58,720,256 | 1,484,836,864 |
| model.layers.6.post_attention_layernorm | 4,096 | 1,484,840,960 |
| model.layers.6.self_attn.k_proj | 4,194,304 | 1,489,035,264 |
| model.layers.6.self_attn.o_proj | 16,777,216 | 1,505,812,480 |
| model.layers.6.self_attn.q_proj | 16,777,216 | 1,522,589,696 |